SHYAM KUMAR

AI Security & AI Governance | Applied Cryptography | NIST AI RMF · OWASP LLM Top 10 · MITRE ATLAS
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SUMMARY

CSE undergraduate working on AI security and governance: mapping real AI incidents across OWASP, MITRE ATLAS and the NIST AI RMF, and building the controls that hold when systems fail, from an incident register whose database refuses to rewrite history to a face sign-in that never uploads a face. Every project is public and tested.

AI SECURITY, GOVERNANCE & SYSTEMS PROJECTS

Pathivu | AI incident register (Java, MySQL, JDBC) — *Oct 2026*

- Maps each AI incident to OWASP LLM Top 10, MITRE ATLAS and NIST AI RMF and runs the CERT-In 6-hour and DPDP reporting clocks from the moment of awareness
- 36 MySQL triggers and guard procedures make history tamper-evident (hash-chained audit, no DELETE grant); 110 tests, 65 of them attacks on the database

Mugam | Face sign-in where the face never leaves the device (Java, WebCrypto) — *Oct 2026*

- On-device face match and two-move liveness unlock a non-extractable P-256 key that signs a one-time, origin-bound challenge; the server stores public keys only
- Cancelable templates (secret orthonormal rotation + AES-GCM), HMAC-chained audit log, recovery codes as PBKDF2 hashes; 79 tests incl. a still-photo attack in a real browser

Kaaval | Permission layer for AI coding agents (Python) — *Oct 2026*

- Decides allow / ask / deny before an agent's shell, SQL or tool call runs; replays 6 real no-adversary incidents and refuses each; tamper-evident log; 39 tests, zero dependencies

ai-incident-atlas & **NIST AI RMF assessment** | AI incident analysis — *Sep 2026 – Present*

- Mapped 10 real AI incidents across OWASP, ATLAS (v5.6.0) and the AI RMF, each with the one control that would have stopped it; showed 6 agent data-loss incidents had no adversary
- Rated "AI coding agents with production access" High: 7-risk register, 8 owned recommendations, DPDP Rules 2025 / CERT-In analysis

Kaappu | TOTP authenticator from RFC 4226 / 6238 (Java) — *Oct 2026*

- AES-256-GCM vault (PBKDF2, 600k), Shamir secret-sharing backup of the vault key, replay-safe verifier; 70 tests incl. every RFC vector; written threat model

Stewards' Room | Open register of F1 stewards' decisions (Python) — *Oct 2026*

- Parses FIA decision documents while keeping each field's exact source text, runs penalty-points clocks, hash-chains every fetched PDF; indexed 2,629 documents; 44 tests

FairDoor | Deadline-fair rate limiting, simulation — *Oct 2026*

- In a 300-student shared-IP simulation, per-IP limiting left ~68 students past the deadline (worse than none, ~60); per-student fair queuing left 0

redteam-log | AI red-teaming lab notebook — *Aug 2026 – Present*

- All 10 OWASP LLM Top 10 (2025) classes mapped to ATLAS techniques with garak probes and controls; garak + PyRIT + Ollama lab with a runbook

PRODUCT & HACKATHON PROJECTS

Virasat | PEC Hacks 4.0, Panimalar Engineering College (36-hour national hackathon) — *Aug 2026*

- 1st place, FinTech domain: find and claim unclaimed assets across RBI, EPFO, IRDAI, SEBI and LIC, with rule-based document verification. TypeScript, Drizzle ORM, Vercel

AquaNXT | ASU Venture X 2026 — *Mar 2026 – Present*

- Gravity water filters in three form factors sharing one on-device monitoring assistant that reports only what it measures; REC's finalist in the local round

UpAid | Cintana AI Challenge 2026 (team) — *Aug – Sep 2026*

- Evidence-first financial-aid finder for ASU-alliance campuses: every match cites its source; registry of 34 campuses and 69 schemes with human verification before publishing

LEADERSHIP & EXPERIENCE

Head, Tech & Skills Club | ASU Pathways (REC, Shiv Nadar University, Anurag University) — *Sep 2026 – Present*

- Founding head of a cross-university student club; built the club website, tech-skills-council.github.io

General Secretary | Velammal Bodhi Campus (Student Council): school-wide events and sponsorships — *2023 – 2025*

EDUCATION

B.E. Computer Science and Engineering | Rajalakshmi Engineering College, ASU Pathways cohort — *Aug 2025 – May 2029* | SGPA: 7.5

High School, Mathematics and Biology (PCMB) | Velammal Bodhi Campus — *2023 – 2025* | 75% (First Class)

SKILLS

AI security / governance: OWASP LLM Top 10, MITRE ATLAS, NIST AI RMF, DPDP Act & Rules, CERT-In Directions, threat modelling, prompt injection, garak, PyRIT
Security engineering: WebCrypto, ECDSA P-256, AES-GCM, PBKDF2 / HMAC, TOTP, Shamir secret sharing, hash-chained audit logs, CSP / CSRF / cookie hardening
Programming: Java, Python, SQL (MySQL), JavaScript / TypeScript, C, ESP32 / C++, Git
Languages: English (Cambridge C1 Advanced, score 187, 2024) · Tamil (native) · Hindi (DBHPS Prathamic) · French (studied through Class 10 board exams)

CERTIFICATIONS & COURSES

- Anthropic (10+, Sep 2026): AI Fluency, Building with the Claude API, Model Context Protocol, Agents and Workflows, Prompt Engineering
- Microsoft AI Red Teaming 101 (Oct 2026) · ASU Co-Lecturing, Introduction to Engineering (Mar 2026) · NIST AI RMF (in progress)